AIAA Launch Vehicles and Entry Systems (LVES) Technical Committee — Charter (DRAFT)

| Version | Date | Revisions | By |

|---|---|---|---|

| Draft v0.1 | 2026-08-20 | Initial working-group draft for review; expands current LVES scope to reusable upper stages, ISAM/SM&L, and hypersonic — with explicit interfaces | J. Quinn (Exos Aerospace) |

| Draft v0.2 | 2026-08-25 | Added Membership & Participation provision (§3.a) responding to L. Honda's roster-review question and the low-turnout issue; added Dennis Poulos and John Fuller (Anduril) to the working-group roster; added survey-to-charter note | J. Quinn (Exos Aerospace) |

| Draft v0.3 | 2026-08-25 | Grounded §3 and §3.a in the AIAA norm stated by A. Anilkumar on 8/24 — **annual review, 3-year member term renewable at the pleasure of TC leadership** — and layered the active-member / quorum mechanics on top of that norm rather than as a freelance alternative; updated survey Q11 to match | J. Quinn (Exos Aerospace) |

| Draft v0.4 | 2026-08-25 | Adopted J. Vinson's 8/24 proposed membership paragraph **verbatim** as the base §3.a text (participation + invited-to-chair-subcommittee language + the two Adaptive Structures TC sentences he cited); reordered §3.a to be **opportunity-first** per Vinson — step up committee activities and create participation slots *before* any roster pruning; recast the active-member / quorum mechanics from a standing rule into an **optional tool tied to the September 2027 recruiting cycle**, not a mid-cycle striking device; added the "annual recruiting & appointment is uniform across AIAA TCs — stay aligned" note; wired Vinson's three subcommittee topics into §4; updated survey Q11 to test opportunity-first vs. rule-first | J. Quinn (Exos Aerospace) |

> **Status:** DRAFT for the LVES Charter Working Group (Vinson, Honda, Sponable, Ghimire, Anilkumar, Poulos, Fuller, Quinn). Not an approved AIAA document. Format follows the standard AIAA TC charter (Committee Name → Scope → Governance → Roles & Responsibilities), consistent with current AIAA TAD Policy & Procedures, which require a charter reviewed annually by TC leadership.

---

1. Committee Name

Launch Vehicles and Entry Systems (LVES) Technical Committee.

2. Scope

Core (present anchor — carried from the current LVES charter):

To promote the development and implementation of launch vehicle and entry systems; to foster interaction and integration across technical disciplines; and to assure current, comprehensive situational awareness of commercial and government, national and international R&D efforts. The LVES TC brings together industry, academic, and government experts within the framework and charter of the AIAA, serving as a forum for evaluating the design, development, test, and operation of reliable, affordable, and reusable launch and entry systems.

Core scope explicitly includes:

- Expendable launch vehicles (ELVs) and reusable launch vehicles (RLVs) — full vehicle systems, integration, and operations.
- Upper stages, with an explicit **step toward reusable upper stages** and the technologies that enable them: space mobility & logistics (SM&L), in-space servicing, assembly & manufacturing (ISAM), on-orbit / in-space refueling, and dynamic maneuver.
- Entry, descent, and recovery systems for reusable and returning vehicles.
- Hypersonic **suborbital rockets** (rocket-powered boost/sounding/test platforms), where they share vehicle, propulsion, GNC, thermal, and operations technology with launch and entry.
- **Cross-cutting operability, reliability, and safety** of ELVs, RLVs, and the concepts above — treated as a standing theme, not a sidebar, since it is the connective tissue across every vehicle

class.

Interfaces (coordinate, do not absorb):

The following are named as interface areas so the door stays open to shared technology without diluting the charter or duplicating another committee's mandate:

- **Rocket-powered hypersonic aircraft** and **airbreathing hypersonic** concepts — strong common-technology overlap (structures, TPS, propulsion integration, operability); coordinated with the relevant Hypersonic / High-Speed Air-Breathing Propulsion committees rather than owned here.
- **Propulsion** (Liquid, Solid, Hybrid, Nuclear & Future Flight) — LVES addresses propulsion at the vehicle-integration level; component R&D remains with the propulsion TCs.
- **Space logistics, ISAM, and spacecraft** committees — LVES owns the launch/upper-stage/entry vehicle and its step to reusability; the on-orbit servicing mission architecture is a shared boundary.
- **Space transportation policy, spaceports, range & safety, GNC** — coordinated for standards, operability, and regulatory input.

Explicit exclusions (constrain the problem):

- Payload/spacecraft bus design beyond the launch/upper-stage interface.
- Air-breathing propulsion component development (interface only).
- Purely orbital on-station operations unrelated to launch, upper-stage, entry, or reusability.

3. Governance

Technical Committee under the [AIAA Group — TBD by working group, e.g., Space & Missiles Group]. Charter and roster reviewed **annually** by TC leadership per AIAA TAD Policy & Procedures. Consistent with the AIAA norm (per A. Anilkumar, 8/24), a member serves a term of **up to three years**, with continuation renewed at the discretion of TC leadership. **The annual recruiting and appointment of TC members is uniform across all AIAA Technical Committees, and this charter stays aligned with that process** (per J. Vinson, 8/24). Membership balanced across industry, academia, and government; across the specialties in scope; and across U.S. and international representation.

3.a Membership & Participation

Base provision (proposed by J. Vinson, 8/24 — adopted as the working text). The following paragraph is the group's converging base language. It draws the active-member sentences directly from the AIAA Adaptive Structures TC charter that L. Honda circulated, so the provision matches an approved AIAA precedent rather than a one-off:

> Committee members are expected to actively participate in the committee meetings and bring value to the charter activity of the TC. Committee members will be invited to form and chair subcommittees covering topics of interest to the members. Potential subcommittees may address a wide range of topics within the charter including but not limited to: (1) understanding and forecasting the market for LVESs, (2) fostering better communication and interactions with the contemporary LVES industry, and (3) expanding the educational outreach of the LVES Technical Committee. While it is understood that not all members will contribute equally to the committee, membership implies willingness to serve and contribute to the committee and technical community. If a member is not contributing, the chair may ask them to rotate off the committee.

Sequencing — opportunities first, pruning last (per J. Vinson, 8/24). On the low-turnout problem L. Honda raised (roughly 10 of ~50 at the last meeting), the working-group posture is to **step up committee activity to create more ways to participate before removing anyone from the roster**. The subcommittee slots above are the primary instrument: each named subcommittee (market forecasting, industry engagement, educational outreach — plus any the survey surfaces) is a concrete seat that converts a nominal member into an active one. Roster changes are handled through the normal **annual**

recruiting-and-appointment cycle, and the group will **revisit membership at the September 2027 recruiting window**, not through mid-cycle removals.

Optional participation mechanics (for the group to weigh at the September review — bracketed thresholds for the group to set; survey Q11 will inform them). *These are offered as tools the TC may adopt if, after activities are stepped up, engagement still lags. They are deliberately subordinate to the sequencing above — none of them is a mid-year striking device.*

- **Active-member standard.** A member could be counted as **active** if, over the prior 12 months, they either attend at least **[half]** of scheduled meetings **or** contribute to at least one committee deliverable (subcommittee work, session, roadmap input, white paper, review). This is a way to *read* engagement at the annual review, not to remove anyone between reviews.
- **Corresponding/affiliate status.** Members who meet neither at the annual review could move to a **corresponding/affiliate** status — still on distribution, still welcome to re-engage — freeing a roster slot for the recruiting cycle. Framed as roster hygiene at renewal, consistent with Vinson's "revisit in September" and Anilkumar's three-year term.
- **Quorum.** If the TC wants votes to reflect the working core rather than the full nominal roster, quorum could be defined as **[a simple majority of active members]**. Offered as an option, not a default.
- **Recruiting to fill gaps.** The annual review uses the survey's "gaps" input (who/what discipline is missing) to target new members, so each expiring three-year term and each September cycle becomes an additive recruiting pass, not just pruning.

4. Roles & Responsibilities

4.a Institute (AIAA) activities

- Organize and lead technical sessions, panels, and Forum 360 content at AIAA SciTech, AVIATION, and ASCEND; coordinate co-sponsored sessions with interface committees.
- Develop and maintain a **launch/entry & reusable-upper-stage technology roadmap**; provide standards and policy input where the community needs a common reference.
- Honors & awards oversight; best-practice and lessons-learned capture across ELV/RLV operability, reliability, and safety.

4.b External / community activities

- Situational awareness of commercial and government, national and international R&D — briefed back to the community.
- STEM and workforce outreach; university engagement (e.g., student design, flight-test, and payload opportunities).
- Liaison to interface committees and to external bodies (standards, range/safety, spaceport, policy).

4.c Standing subcommittees (seed set — from J. Vinson's 8/24 proposal, to be confirmed by survey Q10)

These are the participation slots that answer the turnout problem — every member is invited to form or chair one:

1. **Market forecasting** — understanding and forecasting the market for LVESs.

2. **Industry engagement** — fostering better communication and interaction with the contemporary LVES industry.

3. **Educational outreach** — expanding the educational outreach of the LVES TC.

4. **Reusable upper stages / SM&L** and **Operability–reliability–safety** — carried from the scope as candidate standing subcommittees (survey Q10 tests these).

4.d Deliverables (next 12–24 months — proposed)

1. Two or more named technical sessions/panels per forum cycle across the core scope.

2. A first-issue technology roadmap covering ELV/RLV + the step to reusable upper stages (SM&L / ISAM / refueling).

3. A standing operability–reliability–safety working thread with an annual state-of-the-field readout.

5. Objectives Ledger (proposed governance mechanism)

Every committee objective — and every subcommittee in §4.c — carries an **owner, deliverable, due date, and quarterly status** on one shared page. This is included in the charter from day one so the committee is measured on tangible output (named sessions, a roadmap, standards input) rather than minutes. Exos has offered to stand up and maintain the ledger for the group. It also does double duty on the turnout problem: it is where "stepping up activities" becomes visible, and at the annual/September review it is the record of who carried an objective — the evidence base for any active-member reading in §3.a.

Prepared by Exos Aerospace (J. Quinn) as a working-group draft. Bracketed items [ ] require group decision. §3.a base language is J. Vinson's 8/24 proposal (Adaptive Structures TC precedent); the optional mechanics are subordinate to the group's opportunity-first, review-in-September posture. Scope, membership thresholds, and deliverables are being tested by the Round-1 working-group survey (2026-08-25), whose questions map 1:1 onto these sections. Source format: AIAA TC charters (e.g., Adaptive Structures, Transformational Flight) + current AIAA LVES TC scope + AIAA TAD Policy & Procedures.